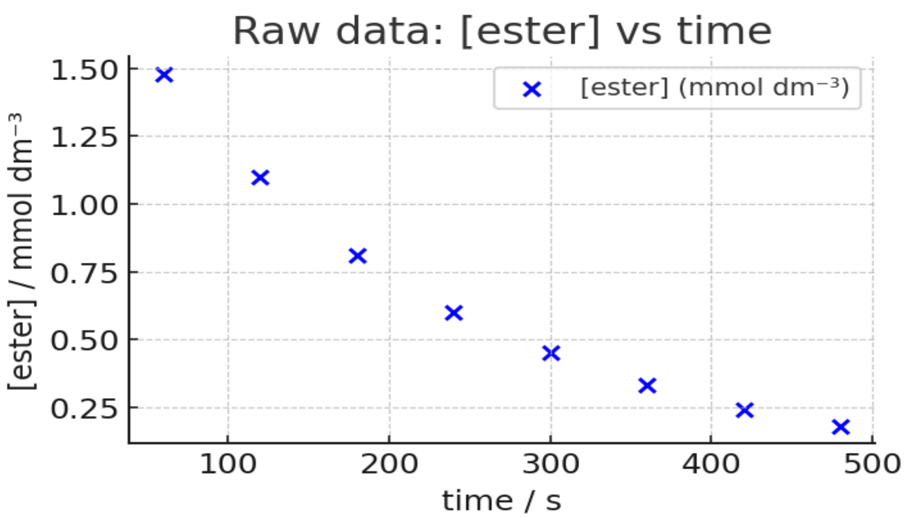
Kinetic Analysis of Ester Hydrolysis

This report analyses the hydrolysis of a 0.002 mol dm■^{3} ester solution with 0.2 mol dm■^{3} NaOH, using time–concentration data. The aim is to determine the order with respect to ester, calculate the half-life and apparent rate constant, and comment on hydroxide dependence.

Experimental data:

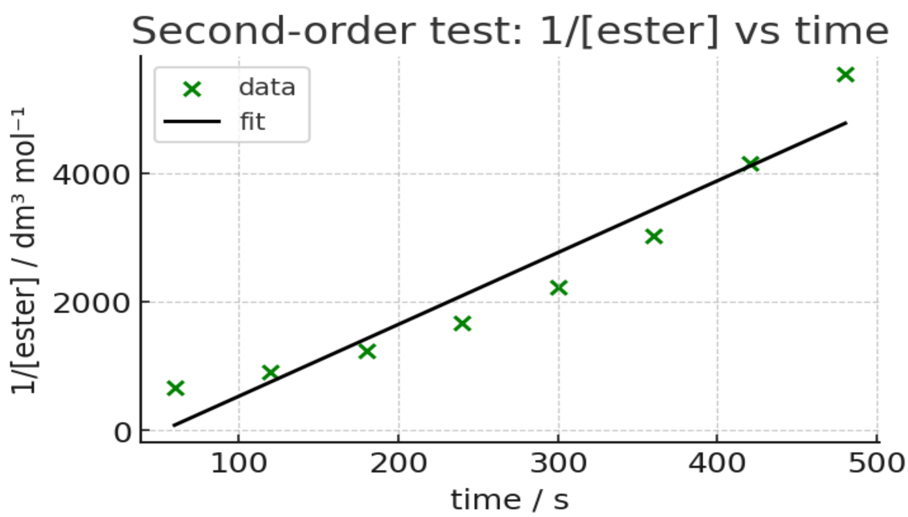
Time / s	[ester] / mmol dm■^{3}
60	1.48
120	1.10
180	0.81
240	0.60
300	0.45
360	0.33
420	0.24
480	0.18

Raw data plot:



First-order test:

Second-order test (for comparison):



Regression Results (first-order test):
ln[ester] = (-0.005028 ± 0.000020)·t + (-6.210584 ± 0.006060)
R^2 = 0.999905

Kinetic Parameters:
Apparent rate constant k = (5.0281e-03 ± 2.0e-05) s^{-1}
Half-life t½ = 137.86 ± 0.55 s

The linearity of ln[ester] vs time confirms that the reaction is first-order with respect to the ester.

Discussion:

a) A plot of ln[ester] vs time gave a straight line, showing the reaction is first-order with respect to ester. This method works because for a first-order reaction, ln[A] decreases linearly with time.

b) From the slope of the line, the apparent rate constant k and the half-life were determined as above.

c) The experiment was carried out under conditions where [OH■] is very large compared to [ester]. Thus [OH■] is effectively constant, so the data only reveal the order with respect to ester, not OH■.

d) To determine the order in OH■, the experiment must be repeated with different initial hydroxide concentrations. Comparing the initial rates (or pseudo-first-order k values) across these experiments would allow deduction of the dependence on [OH■].